CME

Diagnostic Performance of Spurling's Test for the Assessment Subacute and Chronic Cervical Radiculopathy

A Systematic Review and Meta-analysis

Long-Huei Lin, PT, MBA,[1] Ting-Yu Lin, MD,[2] Ke-Vin Chang, PhD, MD,[3,4,5] Chih-Chen Tzang, BS,[6] Wei-Ting Wu, MD,[3,4] and Levent Özçakar, MD[7]

Objective: Spurling's test is commonly used to diagnose cervical radiculopathy (CR); however, its diagnostic accuracy varies across studies. This systematic review and meta-analysis aim to evaluate the diagnostic performance of Spurling's test for cervical radiculopathy.

Design: A systematic search of PubMed, ClinicalTrials.gov, Cochrane Library, and Physiotherapy Database was conducted following Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. Studies assessing Spurling's test against imaging or electrodiagnostic tests were included. A bivariate random-effects model calculated pooled sensitivity, specificity, likelihood ratios, and log diagnostic odds ratios.

Results: Eight studies met the inclusion criteria. The pooled sensitivity of Spurling's test was 0.53 (95% confidence interval: 0.29–0.78), higher with imaging (0.67) than with electrodiagnostic testing (0.31) as the reference standard. Combining neck rotation or extension increased sensitivity to 0.67, compared to 0.31 without. The pooled specificity was 0.92 (95% confidence interval: 0.88–0.96), 0.93 with imaging and 0.91 with electrodiagnostic testing as the reference standard. The pooled positive likelihood ratio was 3.28, negative likelihood ratio was 0.28, and the log diagnostic odds ratio was 2.74.

Conclusions: Spurling's test is highly specific but has low sensitivity for cervical radiculopathy diagnosis, making it better suited for confirmation than screening. Sensitivity may improve with added maneuvers and imaging as a reference.

Key Words: Cervical Root, Physical Examination, Diagnostic Test, Sensitivity, Specificity

(*Am J Phys Med Rehabil* 2025;104:717–723)

Cervical radiculopathy (CR) arises from conditions that affect the nerve roots, leading to symptoms such as pain, numbness, weakness, and paresthesia in the upper limbs. These conditions include compression, irritation, traction, or lesions on the nerve roots, which are often caused by herniated discs, foraminal narrowing, or degenerative/spondylitis changes. The incidence of CR is 3.3 cases per 1000 people, with the highest occurrence seen in individuals in their 40s and 50s.[1] C6 and C7 nerve roots are the most frequently affected, probably because of the fact that intervertebral foramina are larger in the upper than the lower cervical region.[2]

While diagnostic (e.g., myelography, computed tomography or magnetic resonance) imaging or electro-diagnostic evaluation is commonly used to confirm the diagnosis,[2] screening/physical tests can also be contributory. Herein, Spurling's test (ST), also known as the foraminal compression test, is a widely used maneuver for CR. The test is considered positive when radicular pain is reproduced (pain radiates to the shoulder or upper

To Claim CME Credits: Complete the self-assessment activity and evaluation online at http://www.physiatry.org/JournalCME

CME Objectives: Upon completion of this article, the reader should be able to: 1) Understand the diagnostic performance of Spurling's test for cervical radiculopathy. 2) Identify strategies to improve diagnostic performance during clinical testing. 3) Recognize the clinical role of Spurling's Test in the diagnosis and management of cervical radiculopathy.

Level: Advanced

Accreditation: The Association of Academic Physiatrists is accredited by the Accreditation Council for Continuing Medical Education to provide continuing medical education for physicians.

The Association of Academic Physiatrists designates this Journal-based CME activity for a maximum of *1.0 AMA PRA Category 1 Credit(s)*™. Physicians should only claim credit commensurate with the extent of their participation in the activity.

[1]School of Physical Therapy and Graduate Institute of Rehabilitation Science, College of Medicine, Chang Gung University, Taoyuan, Taiwan; [2]Department of Physical Medicine and Rehabilitation, Lo-Hsu Medical Foundation, Inc., Lotung Poh-Ai Hospital, Yilan, Taiwan; [3]Department of Physical Medicine and Rehabilitation, National Taiwan University Hospital, College of Medicine, National Taiwan University, Taipei, Taiwan; [4]Department of Physical Medicine and Rehabilitation, National Taiwan University Hospital, Bei-Hu Branch, Taipei, Taiwan; [5]Center for Regional Anesthesia and Pain Medicine, Wang-Fang Hospital, Taipei Medical University, Taipei, Taiwan; [6]School of Medicine, College of Medicine, National Taiwan University, Taipei, Taiwan; [7]Department of Physical and Rehabilitation Medicine, Hacettepe University Medical School, Ankara, Turkey.

All correspondence should be addressed to: Ke-Vin Chang, PhD, MD, Department of Physical Medicine and Rehabilitation, National Taiwan University Hospital, Bei-Hu Branch and National Taiwan University College of Medicine, Taipei, Taiwan.

Ting-Yu Lin is in training.

Financial disclosure statements have been obtained, and no conflicts of interest have been reported by the authors or by any individuals in control of the content of this article.

Funding: The present study was funded by National Taiwan University Hospital, Bei-Hu Branch (grant nos. 113-BH002 and 113-BH004), the Ministry of Science and Technology (grant nos. MOST 106-2314-B-002-180-MY3, 109-2314-B-002-114-MY3, 111-2314-B-002–161, NSTC 113-2314-B-002-208-MY2 and NSTC 113-2314-B-002-209-MY2), and the Taiwan Society of Ultrasound in Medicine.

Financial benefits to the authors: Financial disclosure statements have been obtained, and no conflicts of interest have been reported by the authors or by any individuals in control of the content of this article.

Data availability statement: The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.ajpmr.com).

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

ISSN: 0894-9115

DOI: 10.1097/PHM.0000000000002707

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

extremity ipsilateral to the direction of neck side bending). Some studies also suggest that this test can be combined with neck extension or rotation to enhance diagnostic accuracy. The underlying mechanism is to reduce the size of the intervertebral foramen, thereby increasing the compression on the exiting nerve roots.[3]

Although several studies have assessed the diagnostic performance of ST in diagnosing CR, their results have been inconsistent. A previous systematic review suggested that ST may have relatively high specificity but not sensitivity.[4] Likewise, this study aims to conduct a systematic review and meta-analysis in order to calculate a pooled effect size to offer stronger statistical evidence on the validity of ST for diagnosing subacute and chronic CR.

METHODS

Search Strategy

The systematic search was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 guidelines.[5] Table S1 provides the Preferred Reporting Items for Systematic Reviews and Meta-Analyses checklist for references (See Supplementary Checklist, http://links.lww.com/PHM/C703). The study protocol was registered on Inplasy.com (INPLASY2024100114). Two authors (L-HL and T-YL) screened the relevant articles within multiple databases, including PubMed, ClinicalTrials.gov, Cochrane Library, and the Physiotherapy Database (PEDro). The PICO question was formulated as follows: Population (P) – patients with CR; Intervention (I) – examination using ST; Comparison (C) – myelography, computed tomography, magnetic resonance imaging, or electrophysiological studies; Outcome (O) – diagnostic performance of ST. The search utilized combinations of the following terms: ("foraminal compression test" OR "Spurling's test" OR "Spurling's maneuver") AND ("cervical radiculopathy" OR "cervical nerve root compression" OR "cervical nerve root impingement" OR "cervical nerve root irritation"). Our literature search covered the period from the inception of these databases up until October 2024, with additional specifics provided in Table S2.

Inclusion/Exclusion Criteria

The inclusion criteria were as follows: (1) participants aged 18 and older; (2) ST as the index test; (3) reference standards including myelogram, computed tomography, magnetic resonance imaging, electromyography, nerve conduction studies, or surgical findings; and (4) cross-sectional or cohort studies with prospective or retrospective data on ST for diagnosing CR.

Exclusion criteria were: (1) case reports, case series, reviews, study protocols, editorials, or commentaries; and (2) studies without sufficient data on true positive, false positive, true negative, or false negative results for the index test.

Data Extraction

Two review authors (L-HL and T-YL) independently extracted data and diagnostic two-by-two tables (true positive, false positive, true negative, and false negative of index test results) for each study. Two-by-two tables were reconstructed using information on relevant parameters (e.g., sensitivity and specificity) if they were not available. In case of disagreement, a final decision was made after discussion with the corresponding author based on criteria ensuring consistency and accuracy between the data presented in the manuscript and the corresponding registry.

Outcome Measurements

The diagnostic performance of ST was assessed by its sensitivity and specificity for diagnosing CR. An ideal test should accurately distinguish between affected and unaffected subjects, so we also used likelihood ratios (LRs) and diagnostic odds ratios (DORs) to measure this distinction. Sensitivity and specificity were classified using a three-point scale: "low" (0.0–0.33), "moderate" (0.34–0.66), and "high" (0.67–1.0).[6] Positive LR below 2 or negative LR above 0.5 indicate limited to no diagnostic value.[7] The value of a DOR ranges from 0 to infinity, with higher values indicating better discriminatory test performance.[8] The DOR value of 1 indicates that the test cannot distinguish between patients with the disorder and those without it.[8] However, because of the potentially large values of the DOR, we also took the logarithm (logDOR) of values to facilitate plotting and calculation.

Quality Assessment

Two independent authors assessed the quality of the included studies using the QUADAS-2 tool,[9] which evaluates four key areas: (1) patient selection, (2) index test, (3) reference standard, and (4) the flow and timing of the primary studies in the meta-analysis. For each area, the risk of bias was categorized as high, low, or unclear. Any differences in the quality assessment were resolved through consensus or determination by the corresponding author. In this meta-analysis, the results of the quality assessment did not determine whether a study was included or excluded. Inclusion or exclusion was based solely on the pre-established criteria.

Statistical Analysis

The diagnostic performance of ST was evaluated by the average sensitivity, specificity, positive/negative LRs, and DOR using a bivariate random-effects model. The variability of the target parameters across the included studies was determined using I^2, which denotes the proportion of variation across studies that is caused by heterogeneity rather than chance. Significant heterogeneity was assumed for an $I^2 > 50\%$.[10] Subgroup analyses were performed based on the type of ST (with or without additional cervical extension/rotation) (Fig. S1) and reference test (diagnostic imaging or electrodiagnostic testing). The original ST involves lateral bending combined with compression. However, subsequent studies have explored variations of the test that incorporate cervical extension or rotation. The modifications might impact the diagnostic performance of the test. On the other hand, the different reference tests could influence diagnostic outcomes because each reference test has varying diagnosis performance on CR. For example, the sensitivity of magnetic resonance imaging and electrodiagnostic testing for diagnosing CR was 93% and 42%, respectively. Although the positive predictive values of magnetic resonance imaging and electrodiagnostic testing were similar, at 91% and 86%, magnetic resonance imaging demonstrated a significantly higher negative predictive value, reaching 25% compared to electrodiagnostic

© 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

testing 7%. These factors led us to select the type of reference test and ST as variables for subgroup analysis.

If a study included both acute and chronic phase subjects, we extracted data from the chronic phase for our analysis. Additionally, if ST was reported with and without added extension/rotation, we prioritized data without these maneuvers to assess the diagnostic performance of the original ST.

To evaluate potential publication bias, we followed the guidelines from the Cochrane Handbook for Systematic Reviews of Interventions. This involved visually inspecting funnel plots and conducting Egger's regression test.[11] Moreover, we also conducted Deeks' funnel plot asymmetry test to address the publication bias which focused on logDOR.[12] Statistical analyses were performed using Stata BE (Stata Statistical Software: Release 18.5, StataCorp LP, College Station, TX), with statistical significance set at $P < 0.05$.

RESULTS

Study Identification and Selection

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses flowchart for the literature search is shown

TABLE 1. Characteristics of the included studies

Author, Year	Country	Study Design	Participants (Female/Male), Age	Symptom Duration, Description	Spurling Test Description	Reference Standard
Viikari-Juntura, 1989	Finland	Prospective cohort study	43 (16/27), 33–80	NA, participants with cervical disc disease such as spondylosis and/or disc herniation	Lateral flexion + rotation + compression	Metrizamide myelography Grade 0: Normal findings; Grade 1: Root amputation only; Grade 2: Filling defect <50% of LCC; Grade 3: Filling defect >50% of LCC, no spinal canal deformation; Grade 4: Spinal canal deformation with total obstruction.
Tong, 2002	USA	Cross-sectional study	255, NA	NA, participants referred for electrodiagnosis of upper extremity nerve disorders	Lateral flexion + extension + compression	EMG Positive sharp waves, fibrillations, reduced recruitment, high-amplitude/long-duration MUAPs, or >30% polyphasic units.
Wainner, 2003	USA	Prospective diagnostic test study	82 (41/41), 45 ± 12	NA, Participants referred to electrophysiologic labs for suspected CR or carpal tunnel syndrome.	Lateral flexion + compression	Needle EMG + NCT CR diagnosis in EDX: normal upper limb NCS ± late response issues, spontaneous activity, neurogenic MUAPs, reduced recruitment in myotomal pattern.
Shah, 2004	India	Prospective cohort study	50 (13/37), 22–60 yrs	2 wks–36 mos, participants with neck and arm pain suggestive of radicular pain	Extension + lateral flexion + compression	MRI, or surgical Findings Root canal diameter at disc prolapse was measured on T2W axial MR images and compared to the unaffected side.
Shabat, 2012	Israel	Cohort study	257 (98/159), 19–78 yrs	lasting ≥4 wks, participants who had symptoms of unilateral CR	Extension + rotation + compression	CT or MRI Imaging results were categorized into three groups: no disc lesion/stenosis, unilateral disc herniation/stenosis, or bilateral disc herniation/stenosis.
Ghasemi, 2013	Iran	Cross-sectional study	97 (72/25), 46.32 ± 13.97	≥3 wks, participants had neck and radicular pain	Lateral flexion + compression	EMG With neurogenic MUAP with or without spontaneous muscle activity
Park, 2017	Korea	Cross-sectional study	135 (57/78), 25–80 yrs	NA, participants referred to pain clinic with a history neck pain	Extension + lateral flexion + compression	MRI (no detail descriptions of criteria)
Sleijser-Koehorst, 2021	Australia	Cross-sectional study	134 (65/69), 49.9 ± 10.7	26 (median) wks, participants referred to the clinic presenting with neck or arm pain.	Extension + rotation lateral flexion +/− compression	MRI (no detail descriptions of criteria)

CT, computed tomography; EMG, electromyography; LCC, lateral contrast column; MRI, magnetic resonance imaging; MUAP, motor unit action potential; NCT, nerve conduction test; wks, weeks; mos, months; yrs, years; NA, not available.

© 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

TABLE 2. The methodological quality of the included studies assessed by QUADAS-2

	Risk of Bias				Applicability Concerns		
Study	**Patient Selection**	**Index Test**	**Reference Standard**	**Flow and Timing**	**Patient Selection**	**Index Test**	**Reference Standard**
Viikari-Juntura, 1989	Low	Low	Low	High	Low	Low	Low
Tong, 2002	Low	Low	Low	Low	Low	Low	Low
Wainner, 2003	Low	Low	Low	Low	Low	Low	Low
Shah, 2004	Low	Low	Low	High	Low	Low	Low
Shabat, 2012	Low	Low	Low	High	Low	Low	Low
Ghasemi, 2013	Low	Low	Low	Low	Low	Low	Low
Park, 2017	Low	Low	Low	Low	Low	Low	Low
Sleijser-Koehorst, 2021	Low	Low	Low	Low	Low	Low	Low

in Figure S2. A total of 520 articles were initially accessed from the electronic databases. After reading their full texts, eight articles met the inclusion criteria and were enrolled in the meta-analysis.[13–20] The reference tests were different in these articles. One used metrizamide myelography,[13] two used magnetic resonance imaging,[19,20] one used myelogram computed tomography or magnetic resonance imaging,[17] one used magnetic resonance imaging or surgical finding,[16] two used electromyography,[14,20] one used electromyography combined with nerve conduction studies.[15] Characteristics of the included studies are listed in Table 1 and the reasons for article exclusion are summarized in Table S3.

Methodological Quality of the Included Studies

Table 2 illustrates the methodological analysis of the included studies based on QUADAS-2. All studies showed a low risk of bias regarding the domains of patient selection, index test, and reference standard. Three studies showed a high risk of bias in the domain of flow and timing.[13,16,17] Two studies used different reference tests for some patients.[16,17] One study had a high rate of patient loss to follow-up, and not all included patients underwent the same reference standard or index test.[13] All studies showed low concern regarding applicability.

Outcome Measurement

Two by two contingency table was showed in Table S4. The pooled sensitivity of ST was 0.53 (95% confidence interval [CI]: 0.29–0.78; I^2 = 98%; Figure 1). Subgroup analyses showed that sensitivity varied based on the ST with/without added neck rotation/extension and the type of reference test. Specifically, ST demonstrated pooled sensitivity of 0.61 (95% CI: 0.38–0.84; I^2 = 97.1%) when combined with neck rotation/extension, whereas it was at 0.32 (95% CI: −0.03–0.67; I^2 = 92.3%) without these maneuvers (Fig. 2A). Additionally, when diagnostic imaging was used as the reference test, the pooled sensitivity increased to 0.67 (95% CI: 0.43–0.90; I^2 = 97.1%), compared to 0.31 (95% CI: 0.09–0.54; I^2 = 84.6%) with electro-diagnostic testing (Fig. 2B).

The pooled specificity of ST across all the included studies was 0.92 (95% CI: 0.88–0.96; I^2 = 63.9%; Figure 3). Subgroup analyses showed that ST showed a specificity of 0.93 (95% CI: 0.89–0.97; I^2 = 67%) when combined with neck rotation/extension, compared to 0.86 (95% CI: 0.77–0.95; I^2 = 0.0%) when these maneuvers were excluded (Fig. 4A). The pooled specificity was 0.93 (95% CI: 0.86–0.99; I^2 = 70.0%) when diagnostic imaging was used as the reference test, and 0.91 (95% CI: 0.87–0.96; I^2 = 10.6%) when electrodiagnostic testing was used (Fig. 4B).

The pooled positive LR, negative LR, and log DOR were as follows: 3.28 (95% CI: 1.46–5.10; I^2 = 45.70%), 0.28 (95% CI: 0.09–0.47; I^2 = 70.8%), and 2.74 (95% CI: 1.49–3.99; I^2 = 84.4%) (Figs. S3–S5). The subgroup analysis for these variables is presented in Figures S6–S11. The positive LR and logDOR were potentially higher while the negative LR was potentially lower—in both the diagnostic imaging subgroup and the subgroup using ST combined with neck rotation or extension.

Publication bias

The funnel plots for sensitivity, specificity, positive LR and negative LR displayed asymmetrical distributions of effect

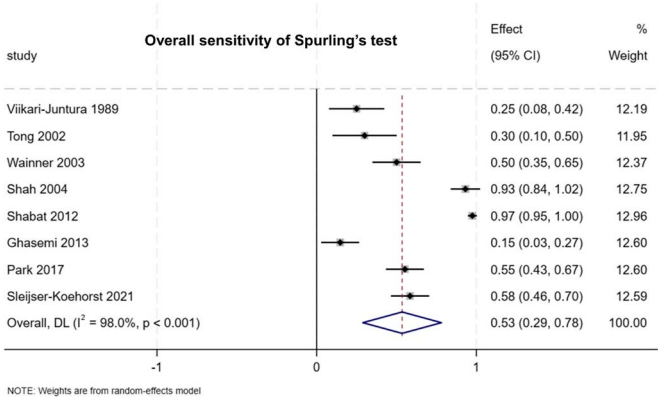

FIGURE 1. Forest plot showing the pooled sensitivity.

© 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

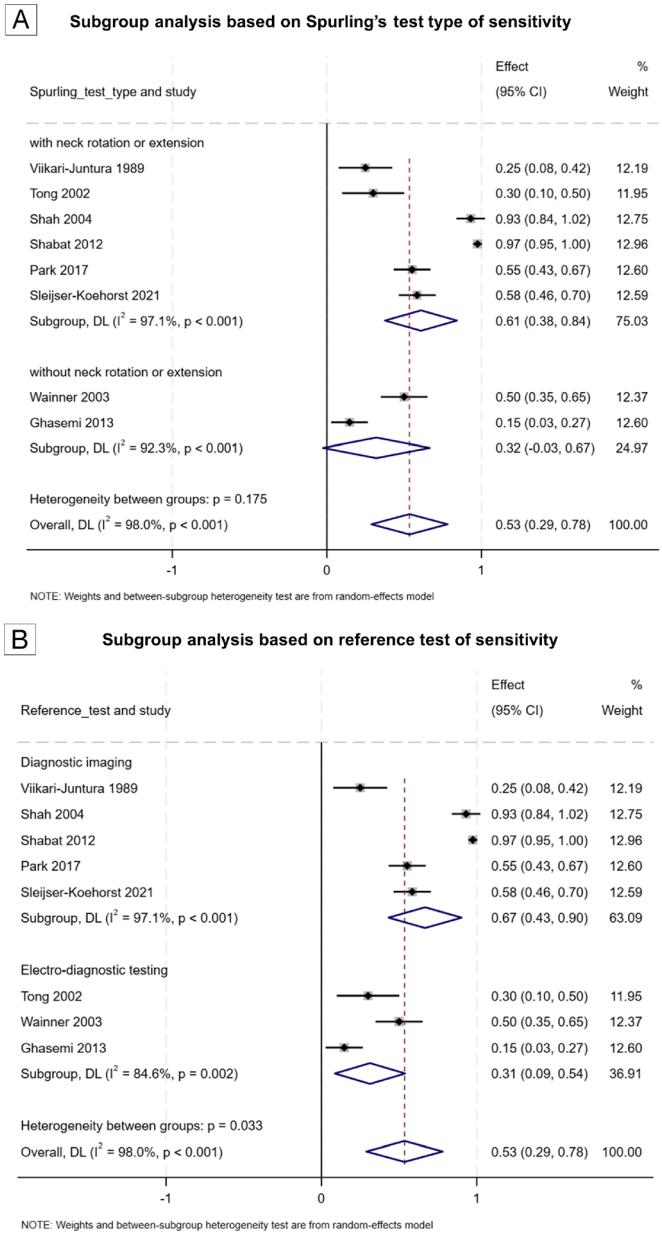

FIGURE 2. Forest plot of the sensitivity subgroup analysis based on (A) Spurling test type and (B) reference test.

sizes (Fig. S1–S16). Egger's regression test indicated significant asymmetry for sensitivity, specificity, positive LR, and negative LR, with *P* values of 0.004, 0.032, 0.019, and 0.003, respectively. In contrast, the funnel plot for logDOR showed a symmetrical distribution of effect sizes, with a nonsignificant *P* value of 0.358. Moreover, the logDOR from Deeks' funnel plot asymmetry test showed no significant evidence of publication bias ($P = 0.246$) (Fig. S17).

DISCUSSION

This meta-analysis showed that ST had high specificity but relatively low sensitivity for the diagnosis of CR. Additionally, the positive and negative likelihood ratios may be crucial for ruling the disorder in or out.[21] Moreover, pooled logDOR indicated that the test could distinguish between patients with

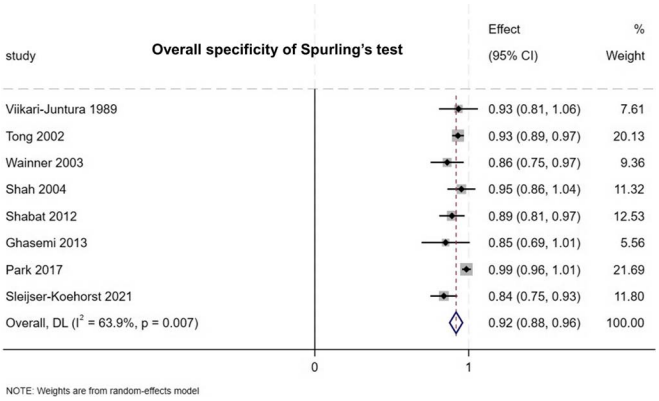

FIGURE 3. Forest plot showing the pooled specificity.

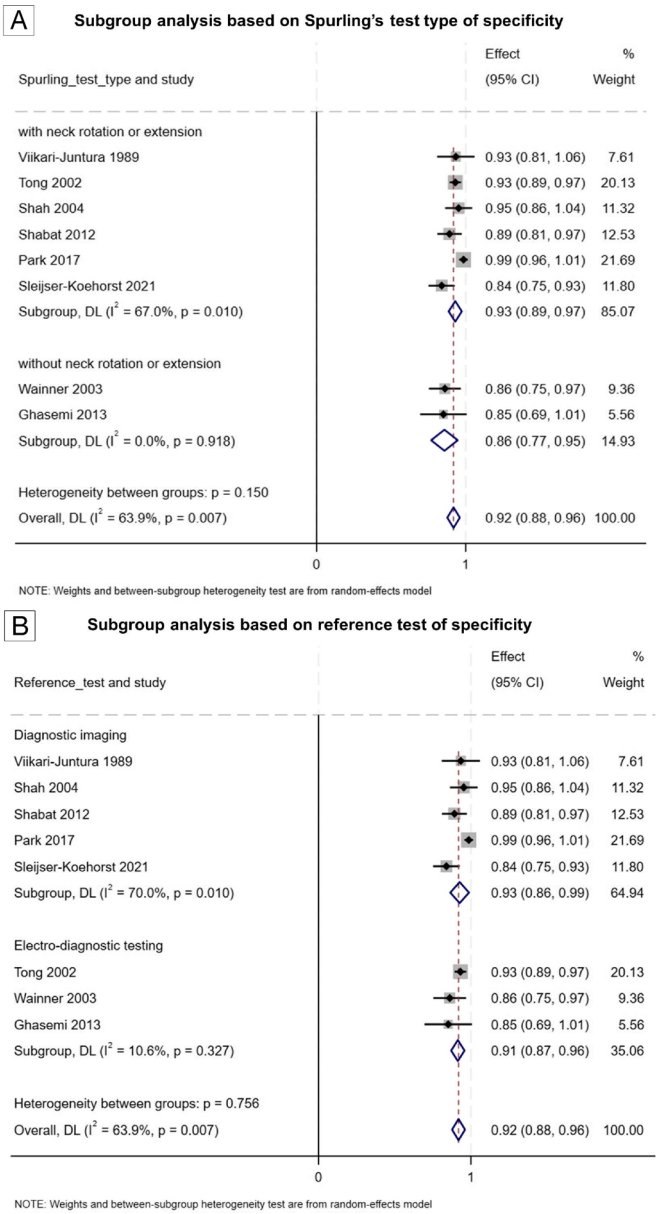

FIGURE 4. Forest plot of the specificity subgroup analysis based on (A) Spurling test type and (B) reference test.

© 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

and without disease. Lower pooled sensitivity was likely observed when the reference tests included electro-diagnostic testing and ST without neck extension/rotation.

This might be the first meta-analysis to evaluate the diagnostic performance of ST to distinguish CR. Rubinstein et al.[22] conducted a systematic review assessing the diagnostic accuracy of provocative tests for diagnosing CR. It comprised six studies, four of which reported sensitivity and specificity for ST, but did not provide additional diagnostic performance metrics, such as the DOR. Thoomes et al.[23] performed a systematic review assessing the value of physical tests for diagnosing CR. This review included five studies three of which involved ST. The authors concluded that ST demonstrated high specificity, ranging from 0.89 to 1.00, while sensitivity varied from 0.38 to 0.97. Lemeunier et al.[4] executed a systematic review assessing the reliability and validity of clinical tests of the cervical spine in adults, including only one study regarding the validity of ST. Therefore, we can imply that our meta-analysis is highly novel in addressing a gap in the current literature.

Our results indicate that ST is characterized by high specificity and relatively low sensitivity. A test with high specificity but low sensitivity effectively identifies individuals without the condition (true negatives) but may miss some cases, leading to false negatives. Clinically, a test with high specificity is useful for confirming a diagnosis if a patient is positive for that test,[24] and it is very likely that they have the condition (few false positives). However, because of low sensitivity, the test may miss some cases, leading to false negatives. This implies that some individuals with the condition (CR in our study) may receive a negative test result, making the test less suitable for initially ruling out the disease.[24] It is better suited as a confirmatory test after an initial screening test with high sensitivity has been used. Targeted movement sequences provoke the nerve's mechanosensitivity. The upper nerve tension test evaluates the nervous system by progressively applying and releasing tension on targeted neural structures. The most commonly performed variation targets the median nerve. In this test, the patient lies supine while the examiner abducts the shoulder to 90 degrees, supinates the forearm, extends the wrist and fingers, and gradually extends the elbow while externally rotating the shoulder. To further increase tension, the head may be laterally flexed away from the tested side. Three key criteria were emphasized during testing[25]: (1) reproduction of neurogenic symptoms such as burning, lightning-like pain, tingling in the neck and arm, or the patient's typical symptoms; (2) differences in range of motion or painful radiation between sides; and (3) symptom changes with structural differentiation, including movements of the cervical spine. Apelby-Albrecht et al.[26] demonstrated that the upper nerve tension test, which has a high sensitivity of 0.97, is effective for diagnosing neurogenic conditions affecting the upper limbs with or without neck pain as it applies mechanical stress to the brachial plexus, spinal nerves or nerve roots and could serve as an initial screening tool when combined with the ST in clinical practice. Moreover, even though the logDOR of ST reaches a level that could distinguish between patients with and without the disease, the logDOR could only serve as a general indicator of diagnostic performance and cannot be used alone to assess the specific effectiveness of a test.[8] There is a close relationship between the DOR and LRs, as DOR is calculated by dividing the positive LR by the negative LR. The pooled positive and negative LRs for ST were 3.28 and 0.28, respectively. Gallagher et al.[27] indicated that positive LRs of 2–10 or negative LRs of 0.5–0.1 are clinically significant, and the values for ST fall within the described range.

Based on our results, we found significant heterogeneity in the diagnostic performance of ST for CR. To investigate further, we first conducted a subgroup analysis based on the type of ST. Tests involving extension/rotation appeared to have higher pooled sensitivity compared to those without these movements. This may be explained by changes in neuroforaminal size during extension/rotation. Farmer et al.[28] showed increased neuroforaminal pressure and nerve root compression as a result of additional neck extension. Recently, Muhle et al.[29] demonstrated that during flexion, the neuroforaminal size increased by up to 31% compared to the neutral position. Conversely, during extension, a reduction in neuroforaminal size of up to 20% was observed. Additionally, at 20° and 40° of ipsilateral head rotation, the neuroforaminal size decreased by 15%–23%, respectively, compared to the neutral position. Mao et al.[30] showed similar results about dimensional changes of the neuroforamina. Consequently, it can be inferred that incorporating neck extension and rotation into the original ST may potentially enhance its diagnostic sensitivity in eliciting CR symptoms. If ST is intended as a screening test, it is essential to include neck extension/rotation to enhance its sensitivity.[31]

In the subgroup analysis based on the type of reference test, we observed that the pooled sensitivity varied between groups. Previous studies have shown that the correlation between magnetic resonance imaging and electrodiagnostic findings in CR was not significant ($P = 0.976$),[32] and the agreement in between for diagnosing CR was only fair (kappa: 0.215–0.342).[33] This may be explained by the following factors. First, timing plays a critical role in electrodiagnostic examinations. For instance, spontaneous activity—such as positive sharp waves and fibrillations—typically appears in the paraspinal muscles within 7–10 days after nerve root compression and extends to limb myotomes within 2–3 wks. Reinnervation findings emerge between 3–6 mos. Negative results might be observed if the testing is performed before denervation occurs, after the denervation findings disappear, or if reinnervation has not yet taken place.[34] Second, the compression of the nerve root can also affect the results of electrodiagnostic testing. In nerve conduction examinations, sensory nerve action potentials typically remain normal unless the dorsal root ganglion at the distal end of the neural foramen is involved.[35] Moreover, if the isolated preganglionic sensory root is injured, resulting in pain and paresthesia, the tests may be negative.[36] Third, the degree of nerve root injury can also impact the outcome. Decreased nerve conduction velocity and prolonged motor latency are typically not observed in cervical radiculopathies unless there is extreme demyelination of the axons.[35]

There are several limitations of this meta-analysis that affect interpretation. First, ST results may vary based on different causes of CR. For example, in patients with a herniated disc, including cervical extension in the test might actually relieve the symptoms.[37] As such, future research should focus on analyzing cases with similar underlying causes. Second, the severity of nerve root compression could influence the results. In the study by Shah et al.[16]—also included in our meta-analysis—participants had failed conservative therapy and were advised

© 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

to undergo surgery. This suggests that the severity of nerve root compression in that study was likely to be greater than in other studies, and this may explain the higher sensitivity observed (making it easier to detect using ST). Thirdly, using magnetic resonance imaging to diagnose CR lies in its inability to consistently correlate imaging findings with clinical symptoms.[38] While magnetic resonance imaging is excellent for visualizing structural abnormalities such as disc herniation, foraminal stenosis, or nerve root compression, these findings are not always indicative of symptomatic radiculopathy. Asymptomatic individuals may also exhibit similar abnormalities on magnetic resonance imaging, leading to potential overdiagnosis. Furthermore, magnetic resonance imaging does not assess dynamic factors, such as positional nerve compression, and may not capture the functional impact of the pathology.[39] Metrizamide myelography has limitations in diagnosing CR, primarily due to its lower sensitivity in visualizing soft tissue structures like nerve roots and intervertebral discs compared to magnetic resonance imaging.[40] While metrizamide myelography is effective in detecting bony abnormalities such as foraminal narrowing or osteophytes, it provides limited information about disc herniation or nerve root compression caused by soft tissues. Additionally, metrizamide myelography involves exposure to ionizing radiation, which may be a concern for certain patients, particularly with repeated imaging. Lastly, our meta-analysis revealed publication bias across all pooled outcomes. An updated meta-analysis may be needed to determine whether the current publication bias is affecting the diagnostic performance of ST.

CONCLUSIONS

ST is highly specific but has relatively low sensitivity for diagnosing CR. When positive, it is useful for confirming CR; however, it may miss some cases and therefore may be less suitable for initial screening. Adding neck rotation/extension can improve sensitivity, and using diagnostic imaging (rather than electrodiagnostic testing) as a reference test offers better sensitivity. In short, ST seems to be best utilized as a confirmatory physical test, after initial screening.

REFERENCES

1. Wainner RS, Gill H: Diagnosis and nonoperative management of cervical radiculopathy. *J Orthop Sports Phys Ther* 2000;30:728–44
2. Kuijper B, Tans JT, Schimsheimer RJ, et al: Degenerative cervical radiculopathy: diagnosis and conservative treatment. A review. *Eur J Neurol* 2009;16:15–20
3. Magee DJ, Manske RC: Orthopedic Physical Assessment. Elsevier; 2020: Chapter 3, p. 202
4. Lemeunier N, da Silva-Oolup S, Chow N, et al: Reliability and validity of clinical tests to assess the anatomical integrity of the cervical spine in adults with neck pain and its associated disorders: part 1—a systematic review from the Cervical Assessment and Diagnosis Research Evaluation (CADRE) collaboration. *Eur Spine J* 2017;26:2225–41
5. Page MJ, McKenzie JE, Bossuyt PM, et al: The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71
6. Portney LG, Watkins MP: *Foundations of Clinical Research: Applications to Practice*. Vol 892: Pearson/Prentice Hall Upper Saddle River, NJ; 2009
7. Jaeschke R, Guyatt GH, Sackett DL, et al: Users' guides to the medical literature. III. How to use an article about a diagnostic test. B. What are the results and will they help me in caring for my patients? The Evidence-Based Medicine Working Group. *JAMA* 1994;271:703–7
8. Glas AS, Lijmer JG, Prins MH, et al: The diagnostic odds ratio: a single indicator of test performance. *J Clin Epidemiol* 2003;56:1129–35
9. Whiting PF, Rutjes AW, Westwood ME, et al: QUADAS-2: a revised tool for the quality assessment of diagnostic accuracy studies. *Ann Intern Med* 2011;155:529–36
10. Borenstein M, Hedges LV, Higgins JP, et al: A basic introduction to fixed-effect and random-effects models for meta-analysis. *Res Synth Methods* 2010;1:97–111
11. Page MJ, Higgins JP, Sterne JA: Assessing risk of bias due to missing results in a synthesis. *Cochrane handbook for systematic reviews of interventions* 2019349–74
12. Deeks JJ, Macaskill P, Irwig L: The performance of tests of publication bias and other sample size effects in systematic reviews of diagnostic test accuracy was assessed. *J Clin Epidemiol* 2005;58:882–93
13. Viikari-Juntura E, Porras M, Laasonen E: Validity of clinical tests in the diagnosis of root compression in cervical disc disease. *Spine* 1989;14:253–7
14. Tong HC, Haig AJ, Yamakawa K: The Spurling test and cervical radiculopathy. *Spine (Phila Pa 1976)* 2002;27:156–9
15. Wainner RS, Fritz JM, Irrgang JJ, et al: Reliability and diagnostic accuracy of the clinical examination and patient self-report measures for cervical radiculopathy. *Spine* 2003;28:52–62
16. Shah KC, Rajshekhar V: Reliability of diagnosis of soft cervical disc prolapse using Spurling's test. *Br J Neurosurg* 2004;18:480–3
17. Shabat S, Leitner Y, David R, et al: The correlation between Spurling test and imaging studies in detecting cervical radiculopathy. *J Neuroimaging* 2012;22:375–8
18. Ghasemi M, Golabchi K, Mousavi SA, et al: The value of provocative tests in diagnosis of cervical radiculopathy. *J Res Med Sci* 2013;18(suppl 1):S35–8
19. Park J, Park WY, Hong S, et al: Diagnostic accuracy of the neck tornado test as a new screening test in cervical radiculopathy. *Int J Med Sci* 2017;14:662–7
20. Sleijser-Koehorst MLS, Coppieters MW, Epping R, et al: Diagnostic accuracy of patient interview items and clinical tests for cervical radiculopathy. *Physiotherapy* 2021;111:74–82
21. Portney LG: *Foundations of Clinical Research: Applications to Evidence-Based Practice*. FA Davis, 2020
22. Rubinstein SM, Pool JJM, Van Tulder MW, et al: A systematic review of the diagnostic accuracy of provocative tests of the neck for diagnosing cervical radiculopathy. *Eur Spine J* 2007;16:307–19
23. Thoomes EJ, van Geest S, van der Windt DA, et al: Value of physical tests in diagnosing cervical radiculopathy: a systematic review. *Spine J* 2018;18:179–89
24. Deeks JJ: Systematic reviews of evaluations of diagnostic and screening tests. *BMJ* 2001;323:157–62
25. Grondin F, Cook C, Hall T, et al: Diagnostic accuracy of upper limb neurodynamic tests in the diagnosis of cervical radiculopathy. *Musculoskelet Sci Pract* 2021;55:102427
26. Apelby-Albrecht M, Andersson L, Kleiva IW, et al: Concordance of upper limb neurodynamic tests with medical examination and magnetic resonance imaging in patients with cervical radiculopathy: a diagnostic cohort study. *J Manipulative Physiol Ther* 2013;36:626–32
27. Gallagher EJ: Clinical utility of likelihood ratios. *Ann Emerg Med* 1998;31:391–7
28. Farmer JC, Wisneski RJ: An analysis of neuroforaminal pressures with varying head and arm positions. *Spine* 1994;19:1850–5
29. Muhle C, Resnick D, Ahn JM, et al: In vivo changes in the neuroforaminal size at flexion-extension and axial rotation of the cervical spine in healthy persons examined using kinematic magnetic resonance imaging. *Spine* 2001;26:e287–93
30. Mao H, Driscoll SJ, Li J-S, et al: Dimensional changes of the neuroforamina in subaxial cervical spine during in vivo dynamic flexion-extension. *Spine J* 2016;16:540–6
31. Ashkan K, Johnston P, Moore AJ: A comparison of magnetic resonance imaging and neurophysiological studies in the assessment of cervical radiculopathy. *Br J Neurosurg* 2002;16:146–8
32. Arslan Y, Yaşar E, Zorlu Y: Correlation of electromyography and magnetic resonance imaging findings in the diagnosis of suspected radiculopathy. *Turk J Neurol* 2016;22:055–9
33. Bölük E, Uzunoğlu C, Seçil Y, et al: The role of electrophysiology in the diagnosis of radiculopathy and its comparison with magnetic resonance imaging. *Neurol Sci Neurophysiol* 2021;38:151–7
34. İnal EE, Eser F, Aktekin LA, et al: Comparison of clinical and electrophysiological findings in patients with suspected radiculopathies. *J Back Musculoskelet Rehabil* 2013;26:169–73
35. Pawar S, Kashikar A, Shende V, et al: The study of diagnostic efficacy of nerve conduction study parameters in cervical radiculopathy. *J Clin Diagn Res* 2013;7:2680–2
36. Charles JA, Souayah N: EMG/NCS in the evaluation of spine trauma with radicular symptoms. *Neurol Clin Pract* 2013;3:8–14
37. Kim Y-H, Kim S-I, Park S, et al: Effects of cervical extension on deformation of intervertebral disk and migration of nucleus pulposus. *PM R* 2017;9:329–38
38. Coronado R, Hudson B, Sheets C, et al: Correlation of magnetic resonance imaging findings and reported symptoms in patients with chronic cervical dysfunction. *J Man Manip Ther* 2009;17:148–53
39. Weishaupt D, Schmid MR, Zanetti M, et al: Positional MR imaging of the lumbar spine: does it demonstrate nerve root compromise not visible at conventional MR imaging? *Radiology* 2000;215:247–53
40. Boyd WR, Gardiner GA Jr.: Metrizamide myelography. *AJR Am J Roentgenol* 1977;129:481–4

© 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.